Meg Montañez Davenport

BLOCKCHAIN HEALTH SYSTEMS ENGINEER
WEB3 ARCHITECT · APPLIED RESEARCHER

CONTACT

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BLOCKCHAIN & WEB3

Solidity · Hardhat · Remix IDE · ERC-20/721 · wagmi · viem · MetaMask · DIDs / SSI · IPFS · Filecoin · Merkle DAGs · ZK Proofs · libp2p · Blockscout

ENGINEERING & DATA

Next.js · PostgreSQL 17 · Python · Flask · Streamlit · systemd · REST APIs · Vercel · SSH Ed25519 · Git CI/CD · Agile/Scrum · Tableau

HEALTHCARE & INFORMATICS

HL7 · FHIR · OMOP · HIPAA · EHR Mapping · Clinical Data Models · Orthomolecular

ORGANIZATIONS

- INCOSE — Systems Engineering
- AMIA — Medical Informatics
- HIMSS — Health IT
- ISC² — Cybersecurity
- PDMA — Product Management
- ISOM — Orthomolecular Medicine
- NANP — Nutrition Professionals

CERTIFICATIONS

- **Certified Blockchain Healthcare Professional (CBHP) — Blockchain Council** · Apr 2025
- **Certified Smart Contract Auditor** Blockchain Council · Jan 2026
- **Health Informatics — Johns Hopkins** May 2025
- **Web3 & Blockchain — INSEAD** Apr 2025
- **CISA — Packt** Aug 2025
- **SSI Master — DizmeID** Mar 2025
- **Medical Neuroscience — Duke SOM** Apr 2025

SUMMARY

I design and deploy compliance-aware decentralized systems — using regulated mental healthcare as a proving ground for architectures that scale across complex domains. As founder of Future Systems Lab, I move from concept to production: on-chain deployments, FHIR-aligned consent flows, and governance-aware infrastructure built to HIPAA standards from day one. My background spans naturopathic psychology and blockchain product architecture — I think in systems, across disciplines, at the intersection of governance and user experience. Five Ethereum smart contracts deployed and verified on Sepolia — every FSL system is in production. Pursuing a Doctor of Engineering to formalize applied research methodology and advance systems leadership at the intersection of blockchain, AI, and decentralized governance.

ENGINEERING EXPERIENCE

Lead Engineer — ClaimChain Insurance Claims Governance

2026 — Present

Sepolia: 0xf32979200768e8726d5EC5E5AB0CA7407d64A94e

- Engineered a full PostgreSQL claims lifecycle schema spanning users, patients, providers, visits, claims, documents, audit logs, insurers, eligibility, authorization, and compliance log tables.
- Built Phase 3 SMS-based approval-gated automation with case-insensitive approve/deny/defer, 4-hour timeout, and max-defer logic — deployed to production VPS via systemd.
- Architected ClaimChain as a standalone blockchain governance layer for insurance eligibility verification — integrates with any payment pipeline (Fiat → Stripe → Crypto → MetaMask) and deployable independently of any single platform.

Open-Source Case Study Lead — AlchemistForge / mental-health-on-chain

2026 — Present

- Scaffolded mental-health-on-chain as a five-entry public smart contract awareness series; deployed 24/7 campaign orchestration across Dev.to, Twitter, Discord, GitHub, and Telegram.
- Recorded on-chain case study as Participant #1 with Transmuted and Celebrated events confirmed at wallet 0xf22cbF25deEeA36FFF828BAf73CCb049345eF248.

Founder & Systems Architect — Future Systems Lab

2023 — Present

github.com/Future-Systems-Lab/Future-Systems-Lab-profile

- Deployed and verified five Ethereum smart contracts on Sepolia testnet: HypnoNeuroToken, EncryptHealthToken, MindMasteryNFT, ClaimChain, and AlchemistForge.
- Architected a 12-agent Master Command Center spanning Audit, Smart Contract, Backend Build, Frontend/UI, Compliance, Security, Testing/QA, and Research agents with full dropdown panel orchestration.
- Deployed production VPS automation stack (clawdbot) on OpenClaw VPS: Flask approval server, Telegram bot, Streamlit analytics dashboard, and two systemd services running continuously.
- Engineered EncryptAI — a 19-step canonical monorepo (frontend/, backend/, labs/, contracts/, game/) with MetaMask authentication via wagmi 2.16.9, PostgreSQL 17, and blockchain writes via viem — Step 2 live on Vercel.
- Established SSH-authenticated signed commits, GitHub org governance, and Rights Reserved licensing policy across all FSL repositories.

Healthcare Data Architect — HypnoNeuro Mental Wellness Platform

2023 — Present

HypnoNeuro · EncryptHealth · ClaimChain · NeuroBalance Watch — modular systems engineered to operate independently or as an integrated ecosystem

- Architected a 27-game browser-based therapeutic wellness platform spanning three progressive treatment levels — system design, clinical logic mapping, and token-governed session architecture fully planned; games in active development using React, Web Audio API, and Solidity-governed session tracking.

Health Informatics on FHIR — Georgia Tech Mar 2025

Business Analyst — IBM Jun 2025

Data Analytics · UX Design · PM — Google Jun 2025

IPFS · Filecoin · libp2p · Merkle DAGs IPFS / Filecoin Foundation / ProtoSchool · 2025

- Designed on-chain incentive model linking session completion to token rewards and NFT-based discount tiers, creating verifiable therapeutic progression evidence on-chain.
- Building HIPAA-compliant backend with DID authentication, encrypted communications, and MetaMask wallet-based access as the primary clinical entry point.

Founder & Product Developer — SunCodes Holistic Health

2018 – 2024

- Developed a seed-to-shelf CBD product line in an unregulated market — managed full product lifecycle including formulation, third-party COA testing, GMP documentation, labeling, barcode integration, and retail packaging compliance.
- Led full renovation and interior redesign of a historic downtown smoke shop, converting it into a branded therapeutic wellness center — managed contractor relationships, spatial design, and retail buildout from concept to opening.
- Founded and operated a clinical wellness platform applying naturopathic psychology and orthomolecular frameworks to mental health and trauma recovery, with digital compliance tracking and personalized client protocol systems.

OPEN SOURCE CONTRIBUTIONS

Open Source Contributor — Healthcare & Blockchain

Jul 2025 – Present

Self-Directed / Independent

- **Ethereum / EIPs** — formatting fix in core standards repository
- **OpenZeppelin Contracts** — ReentrancyGuard documentation clarification
- **Uniswap v3-core** — invariant test note for pool mathematics
- **Hyperledger Explorer** — build and runtime notes
- **OpenMRS** — FHIR2 quickstart pointer for healthcare interoperability
- **BigchainDB** — Python quickstart documentation for new developers
- **Balancer v2** — pool math reference documentation
- **Presentation-Exchange** — OIDC4VP consent example for verifiable credential workflows

Applied projects:

- Secure Health Login — practitioner-facing smart wallet authentication app implementing EIP-712, DID/OIDC, and HIPAA-aligned access control — live on Vercel

PUBLICATIONS

> "Ending the Insurance Verification Nightmare"
> America Out Loud · March 1, 2026 · americaoutloud.news
> Argued that blockchain governance resolves systemic insurance fraud — duplicate claims, identity misuse, phantom billing — through decentralized eligibility verification and immutable audit trails.

> "Blockchain Consent Transforms Patient Data Control"
> America Out Loud · March 2026 · americaoutloud.news
> Examined how blockchain-based consent architecture returns data sovereignty to patients — arguing that smart contract-governed consent flows resolve the systemic failures of centralized health data management and establish a verifiable, patient-controlled alternative.

EDUCATION

BS, Management Information Systems — In Progress
CSU Global — Regionally Accredited (HLC)

Doctor of Naturopathic Psychology — Conferred
Kingdom College of Natural Health

PROPOSED APPLIED RESEARCH

> **Decentralized Healthcare Claims Governance:** Engineering a privacy-first, blockchain-secured insurance claims lifecycle system for health providers — advancing ClaimChain from testnet deployment into a rigorous applied research framework examining smart

contract-governed eligibility verification, HIPAA compliance engineering at the contract layer, and interoperability with HL7/FHIR standards at scale. | ClaimChain on Blockscout